

Metehan Altuner

DevOps Team Lead | Platform & Backend Engineer

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PROFESSIONAL SUMMARY

Platform & Backend Engineer with close to four years of experience and end-to-end ownership of a 9-node Red Hat OpenShift cluster running 150+ applications and roughly 1,000 pods. Installed and operates Kafka, Keycloak, 3scale and a clustered GeoServer from scratch, and designed the centralised authentication and endpoint-level authorisation architecture covering 15,000+ users. Builds production event-driven distributed systems in Java/Spring Boot and .NET, and provides technical leadership to a team of three. Works across every layer of the stack, from physical server installation to production code.

SCALE · 9-node OpenShift cluster · **150+** applications · **~1,000** pods · **50+** CI/CD pipelines · **15,000+** users

TECHNICAL SKILLS

PLATFORM & INFRASTRUCTURE

Red Hat OpenShift, Kubernetes, Docker, Linux, physical server installation, resource management (requests/limits), ConfigMap & Secret management, cluster operations

CI/CD & OBSERVABILITY

GitLab CI, Git, automated build–test–deploy pipelines, container image registry, image security scanning, environment standardisation, Elasticsearch, Loki, centralised log aggregation, structured logging

IDENTITY & API MANAGEMENT

Keycloak (realm / role / group design), Red Hat 3scale (products, backends, policies, mapping rules), custom policy development, OAuth2, JWT, API Gateway, endpoint-level authorisation

BACKEND DEVELOPMENT

Java 21, Spring Boot 3.x, Spring Security, Spring Kafka, REST APIs, JPA / Hibernate, .NET 8 / C#, ASP.NET Core, OkHttp

DATA & MESSAGING

Apache Kafka (mTLS, consumer groups, partition assignment, offset management, DLQ), Debezium / CDC, PostgreSQL, MSSQL, Redis, SQL

ARCHITECTURE & DISTRIBUTED SYSTEMS

Microservice architecture, event-driven architecture, idempotency, at-least-once / exactly-once semantics, dead-letter queues, bulkhead isolation, fault tolerance, JVM & GC tuning, Clean Architecture, code review, technical leadership, Agile / Scrum

PROFESSIONAL EXPERIENCE

DevOps Team Lead

Jan 2025 – Present

TCDD Teknik (engineering subsidiary of Turkish State Railways) · Ankara

- Sole owner of a Red Hat OpenShift cluster comprising 3 master (VM), 3 worker (bare metal) and 3 infra (VM) nodes, keeping 150+ frontend and backend applications running at roughly 1,000 pods.
- Installed the 3 bare-metal worker servers and their network connectivity together with the network team, building the infrastructure from the physical layer up.
- Installed Kafka, Keycloak, 3scale and a clustered GeoServer from scratch on the cluster and owns their operational continuity.
- Operates 50+ active GitLab CI pipelines; eliminated manual steps entirely, reaching production automatically within an average of **15 minutes** after a branch merge.
- Runs the centralised observability pipeline that ships environment logs collected via Loki into Elasticsearch, and leads root cause analysis during production incidents.
- Cut per-service memory consumption from **~3 GB to a 150–600 MB range** through pod resource limits, JVM MaxRAMPercentage and .NET GC configuration.
- Provides technical leadership to a team of three, covering code review, technical guidance and backend–platform coordination.

Backend Developer / DevOps Engineer

Nov 2022 – Dec 2024

TCDD Teknik · Ankara

- Developed multiple enterprise microservices in Java / Spring Boot, building REST integrations, service security and data access layers.
- Owned microservice development, API security and OpenShift deployment for enterprise systems supporting railway traffic operations.
- Contributed to the rollout of Keycloak- and 3scale-based API access management; handled data modelling and query optimisation on PostgreSQL and MSSQL.

Software Development Intern

Jul 2022 – Nov 2022

Intern

TCDD (Turkish State Railways), Head Office · Ankara · Supported backend development, database operations and debugging in microservice-based enterprise applications.

SELECTED PROJECTS

Centralised Identity and Endpoint-Level API Authorisation Platform

Keycloak · Red Hat 3scale · Custom Policy · OAuth2 / JWT

- Developed a **custom 3scale policy** binding Keycloak and 3scale together, moving authorisation logic scattered across dozens of backend services entirely into the gateway layer.
- Designed an authorisation model in which each endpoint maps to a role and roles are grouped into policies, bringing **15,000+ users and 27 API products** under a single model.
- Removed the need for backend projects to implement authorisation at all; onboarding a new service became a configuration task rather than a code change.

Distributed Fiscal Receipt (POS) Integration System

Java 21 · Spring Boot 3.5 · .NET 8 · Apache Kafka (mTLS) · OpenShift · MSSQL

- Designed and built an event-driven, polyglot (Java + .NET) architecture around a closed-source, non-thread-safe fiscal printer SDK, reconciling Kafka's at-least-once semantics with the exactly-once requirement of a fiscal receipt.
- Prevented duplicate printing structurally through end-to-end idempotency, pod- and consumer-per-device isolation (bulkhead) and half-print detection; **no duplicate fiscal transactions occur in production**.
- Ensured no receipt is ever lost through durable retry that holds the offset uncommitted, plus a per-device daily cutoff: receipts queue in order while a device is unreachable and print automatically once it returns.
- Eliminated multi-minute blocking on dead sockets via TCP keepalive, half-open connection detection and explicit connect timeouts.
- Also built the enterprise cafeteria and restaurant sales applications the system serves; both remain in production.

Cross-Database CDC Synchronisation Middleware

Debezium · Apache Kafka · Java · Spring Boot

- Set up CDC streams capturing **7 databases** via Debezium and **single-handedly built** the Java middleware that orchestrates the resulting change events.
- Routed inserts, updates and deletes to subscribing databases and tables, removing direct cross-service database coupling and moving synchronisation onto a centralised, event-driven flow.

EDUCATION & LANGUAGES

B.Sc. in Computer Engineering — Kütahya Dumlupınar University

2022

Turkish: Native · **English:** Working proficiency in technical documentation and development